



Percentages and percentage change · Revision notes

Percentages and percentage change: revision notes

GCSE Mathematics · IlluminatED original material · version 1.0 · reviewed 2026-09-30

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MATCHES THE SPECIFICATION

- AQA 8300 · R9 percentages; N12, R16 compound interest

Specification references checked 30 September 2026.

LEARNING OBJECTIVES

- Find a percentage of an amount with and without a calculator
- Use multipliers for percentage increase and decrease
- Calculate percentage change and reverse percentages

EXPLANATION

Per cent means 'out of 100'. 35% is $35/100 = 0.35$. Converting percentages to decimals lets you use a single multiplication.

Without a calculator, build from easy pieces: 10% is $\div 10$, 5% is half of 10%, 1% is $\div 100$. For example 15% of 80 = 10% (8) + 5% (4) = 12.

A multiplier combines the original amount and the change. An increase of 12% means you keep 100% and add 12%, so multiply by 1.12. A decrease of 12% leaves 88%, so multiply by 0.88.

Percentage change = $(\text{change} \div \text{original}) \times 100$. Always divide by the original value, not the new one.

Reverse percentages work backwards from the new value: if a price after a 20% increase is £60, then $1.2 \times \text{original} = 60$, so $\text{original} = 60 \div 1.2 = £50$.

WORKED EXAMPLE

A jacket costs £64. In a sale it is reduced by 15%. What is the sale price?

1. A 15% decrease leaves 85% of the price.
2. Multiplier = 0.85.
3. $£64 \times 0.85 = £54.40$.

Answer: £54.40

KEY TERMS

- Multiplier: The decimal you multiply by to apply a percentage change in one step, e.g. 1.05 for a 5% increase.
- Percentage change: $(\text{Change} \div \text{original value}) \times 100$.
- Reverse percentage: Finding the original amount when you know the value after a percentage change.
- Compound interest: Interest calculated on the original amount plus interest already added; use multiplier to the power of the number of periods.

COMMON MISCONCEPTIONS

Misconception: To undo a 20% increase, take off 20% of the new price.

Correction: Divide by the multiplier (1.2). Taking 20% of the new, larger price removes too much.

Misconception: Percentage change divides by the new value.

Correction: Always divide the change by the original value.

RETRIEVAL QUESTIONS (cover the answers)

1. What multiplier gives a 7% decrease?

Answer: 0.93

2. What is 15% of 80?

Answer: 12

3. A value rises from 40 to 50. What is the percentage increase?

Answer: $10 \div 40 \times 100 = 25\%$